

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

THE EM ALGORITHM IN GENERALITY

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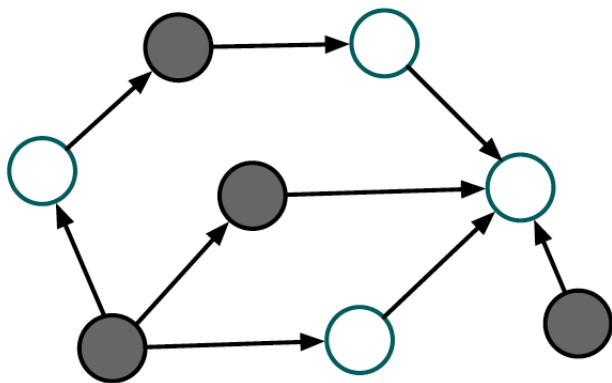
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The Learning Problem

Latent variable model: a graphical model over observable and hidden variables.



Denote:

- X : collection of observable vars
- Z : collection of latent (hidden) vars

- Parametrized family of joint distributions $\{P_{\theta}(x, z) : \theta \in \Theta\}$
- Given observations $x_1, \dots, x_n$, find the θ that maximizes the log-likelihood

$$\sum_{i=1}^n \ln P_{\theta}(x_i) = \sum_{i=1}^n \ln \left(\sum_z P_{\theta}(x_i, z) \right)$$

The EM Approach: Intuition

- Parametrized family of joint distributions $\{P_{\theta}(x, z) : \theta \in \Theta\}$
- Data consists of observations $x_1, \dots, x_n$

Very roughly:

- Choose some initial model
- For $t = 1, 2, \dots$:
- Use $\theta^{(t)}$ to "fill in" the missing z -values:

$$\begin{pmatrix} x_1, _ \\ \vdots \\ x_n, _ \end{pmatrix}$$

- Now fit a model $\theta^{(t+1)}$ to this "completed data"

The EM Algorithm

Let $\hat{P}$ be the empirical distribution: probability mass $1/n$ at each observation.

Let $\theta^{(t)}$ be the parameter-estimate at time t .

- **E-step:** Fill in Z

For each observed x , compute the marginal $P_{\theta^{(t)}}(z|x)$. This gives the “completed joint distribution”

$$\hat{P}^{(t)}(x, z) = \hat{P}(x)P_{\theta^{(t)}}(z|x)$$

- **M-step:** Estimate $\theta^{(t+1)}$

Let $\theta^{(t+1)}$ be the maximum-likelihood fit to the joint distribution $\hat{P}^{(t)}$, i.e.,

$$\arg \max_{\theta} \sum_{x,z} \hat{P}(x)P_{\theta^{(t)}}(z|x) \ln P_{\theta}(x, z)$$

This converges to a local maximum of the log-likelihood.